

Recent BAO–HHT Integration: Files, Results, and Next Steps

This report summarizes the recent development of helper tooling, updates to covariance handling, and key BAO-only and SN+BAO results, plus implications for HHT and recommended next actions.

Executive Summary

- Implemented helper (bao_hht_helper.py) for constant- ρ BAO covariance, residuals/whitening, χ^2 , per- z pairs, and analytic ρ^* .
- BAO-only prefers $\rho^* \approx -0.118 \pm 0.129$; SN+BAO prefers $\rho^* \approx +0.174 \pm 0.064$.
- Using $\rho = +0.143338$ in the combined run already captures $\sim 97\%$ of the attainable χ^2 reduction vs $\rho=0$.
- Whitened BAO residuals are now exportable for HHT in the same style as SNe.

BAO-Only Results (FR, flat; $r_d = 147.090$ Mpc)

ρ	H_0 (km/s/Mpc)	Ω_m	χ^2_{BAO}	Notes
-0.10	68.731	0.29276	59.624	BAO-only min in grid around -0.1
-0.12	68.7337	0.292775	59.603	Consistent with analytic minimum

Quadratic fit near the minimum gives $\rho^* = -0.118 \pm 0.129$ ($\chi^2_{*} \approx 59.60$).

Combined Pantheon+SH0ES + BAO Results (FR, flat; $r_d = 147.090$ Mpc)

Setting	H_0 (km/s/Mpc)	Ω_m	$\chi^2_{\text{SN}} /$ dof _{SN}	χ^2_{BAO}	$\chi^2_{\text{total}} /$ dof _{total}
$\rho=+0.14333$ 8	73.1400	0.235889	2098.521 / 1699 $\approx$ 1.235	236.4175	2334.938 / 1713 $\approx$ 1.363

Analytic optimization at the combined best-fit cosmology gives $\rho^* = +0.174 \pm 0.064$ with $\chi^2_{\text{BAO}}(\rho^*) \approx 236.1908$.

$\Delta\chi^2$ vs $\rho = +0.143338$ is only ≈ 0.227 ; re-running at exactly ρ^* would change posteriors negligibly.

Relation to HHT Analysis

- Exported whitened BAO residuals (using the same covariance as the fit) enable HHT decomposition parallel to SN analysis.

Per- z χ^2 highlights $z \approx 0.845$ and 0.700 as heavy contributors—matching suspect zones in SN residual structure.

- The ρ sign flip (BAO-only negative $\rightarrow$ SN-anchored positive) is expected because ρ de-correlates DM and DH within each pair, and the optimal de-correlation depends on cosmology.

Implications

1) Constant- ρ block covariance adequately summarizes BAO DM–DH cross-correlation for $N=14$.

- Parameter inferences are robust to small mis-specification of ρ near the optimum.
- Whitened BAO residuals align BAO with SN HHT workflows for mode-level comparisons.
- $z \approx 0.7$ – 0.9 remains the primary BAO lever arm—focus HHT inspection here.

Next Steps (Extended Redshift Focus)

- A) Extend BAO coverage: add/update high- z BAO ($\text{Ly}\alpha$ at $z \gtrsim 2$), ensure proper DM/DH pairing and uncertainties.
 - – Introduce $\rho(z)$: two-bin ($z < 0.9$ vs $z \geq 0.9$) or linear slope; compare AIC/BIC to constant ρ .
 - – r_d sensitivity: re-run fit- ρ and combined fits at $r_d \pm \Delta r_d$.
- B) HHT integration: run HHT on exported whitened BAO residuals; compare mode energies/frequencies to SN modes.
 - – Perform leave-one- z -pair-out HHT to gauge influence of localized BAO pairs.
- C) Model checks: try $c_{\text{model}} = \log 1p_z$ vs linear_z ; cautiously add CC or Planck R/r_s to monitor tension migration.
- D) Reproducibility: export `covariance.txt`, `perz_pair_report.csv`, and `whitened_residuals.csv` for each major run.

Paths & Artifacts

- BAO CSV: /Users/boyde/Downloads/bao_MASTER_LONG.csv
- SN data/cov: /Users/boyde/Downloads/Pantheon+SH0ES.dat ;
Pantheon+SH0ES_STAT+SYS.cov
- Helper: ~/.spyder-py3/bao_hht_helper.py
- Cov examples: cov_sweeps/bao_MASTER_cov_blockrho-0.12.txt,
cov_sweeps/bao_MASTER_cov_blockrho0.143338.txt
- Outputs: plamb_runs/COMBINED_BAO_SN_rho_p0.143338,
plamb_runs/tools/bao_hht_at_combined_best/

